Practical No. 17 – Node-RED

Hardware Requirements

1. Raspberry Pi Model 400
2. Power adapter
3. HDMI-to-Micro-HDMI cable
4. Mouse
5. Monitor
6. Push button
7. Resistor
8. LED
9. Jumper wires (female-to-male)

Diagram / Connections

```
GPIO 12 → 5V power (GPIO 2)
GPIO 23 → Resistor → LED anode
LED cathode → GND
```

LED terminals: Anode (+) = longer terminal; Cathode (–) = shorter terminal.

Terminal Commands

```
bash <(curl -sL https://github.com/node-red/linux-installers/releases/latest/download/update-nodejs-and-nodered
```

```
node-red start
node-red stop
node-red restart
node-red reload
```

Node-RED Flow

```
GPIO Input → Switch 1 → Switch 2 → Switch 3
                ↓          ↓
             Change 1    Change 2
                ↓          ↓
            GPIO Output
```

Node Configuration

1. GPIO Input
Select the pin number connected to the push button.

2. Switch 1
Property: `global.state`
Rule 1: `== 0`
Rule 2: `== 1`

3. Switch 2
Property: `msg.payload`
Rule: `== 1`

4. Switch 3
Property: `global.state`
Rule 1: `== 0`
Rule 2: `== 1`

5. Change 1

```
global.state = 1
msg.payload = 1
```

6. Change 2

```
global.state = 0
msg.payload = 0
```

7. GPIO Output

Select the pin number connected to the LED.